Input: $x \in \{0,1\}^n$
Output: $a \in \{0,1\}$

$v \leftarrow$ root of T
$p \leftarrow ?^n$
Sample $\boldsymbol{\rho} \sim R^x_\mu$
while v is not a leaf **do**
    $D^{v,p} \leftarrow \{j \in [n] : p_j = ? \wedge \mathrm{rank}(Q^{\preceq v}_{S(p,\star)+j}) = \mathrm{rank}(Q^{\preceq v}_{S(p,\star)} + 1\}$
    if $D^{v,p} = \emptyset$ **then**
        Infer $b^v \leftarrow \langle x, Q^v \rangle$ from the fact that $(Q^{\prec v})^T x = b^{\prec v}$ and $x_j = 0$ for all $p_j = 0$
    else
        for $j \in D^{v,p}$ **do**
            $p_j \leftarrow \boldsymbol{\rho}_j$
            if $p_j = \star$ **then**
                break
            end if
        end for
        Query $b^v \leftarrow \langle x, Q^v \rangle$
    end if
    Move $v \leftarrow \mathrm{child}(v, b^v)$.
end while
return $L(v)$